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Master Thesis

Energy-Guided Sampling for Controllable Text Generation: From Langevin Dynamics to Amortized Inference

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(Sarthak Singh, Stuttgart)

Abstract

Controllable text generation asks a language model to produce text that satisfies a global property, such as a sentiment, a topic, or a factual constraint, that cannot be evaluated until a sequence is complete. A promising and increasingly popular family of methods approaches this problem without retraining the model: it treats the frozen likelihood of a pretrained autoregressive language model as an *energy function* over sequences and samples from that energy with gradient-guided Markov chain Monte Carlo, most commonly a discretization of Langevin dynamics. The appeal of the framework is that an arbitrary constraint can be added to the energy at inference time, so a single frozen model can, in principle, be steered toward any objective without a new fine-tuning run. This thesis sets out to realize that promise and, in the course of doing so, finds that its central premise does not hold.

We conduct a systematic study across five energy functions, spanning a supervised fine-tuned GPT-2 Large, three GFlowNet-tuned variants of the same base model, and a Llama-3 8B cross-architecture control, comprising one hundred and forty-five sampling configurations on a masked-token recovery benchmark constructed from ROCStories. We evaluate two samplers, a Discrete Langevin Sampler that remains in token space and a Continuous Langevin Sampler that operates in the embedding space, under a factorial combination of Metropolis–Hastings correction, gradient normalization, and annealing schedule length. The headline result is negative and it is consistent: the gradient of a frozen autoregressive likelihood is not a usable search direction on the discrete text manifold. In a controlled ablation that isolates the direction of the gradient from its magnitude, following the exact gradient is statistically indistinguishable from following a random direction of the same norm, on every model tested.

Rather than stopping at this observation, the thesis explains it. We establish three interlocking mechanisms. First, a *linearization failure*: the first-order Taylor surrogate that the discrete sampler uses to rank candidate tokens is essentially uncorrelated with the true change in sequence energy, because the smallest admissible discrete move in the embedding space is far larger than the radius over which the local linear model is valid. Second, a *likelihood trap*: on our own models, the text of highest likelihood is the most degenerate, so any procedure that concentrates probability mass on low-energy regions is optimizing toward poor text by construction; we measure a monotone relationship between per-token likelihood and repetition rate. Third, a *Metropolis–Hastings breakdown* that is specific to the continuous sampler:

because the energy is defined through a nearest-neighbour projection into the vocabulary, its drift is discontinuous at every cell boundary, and we measure that the acceptance ratio is destroyed by the proposal term rather than the target term, so the correction rejects precisely the moves that would change a token.

We then ask whether amortizing the search escapes these mechanisms. Using a GFlowNet, which learns a sampling policy from a scalar reward and never differentiates the energy, we observe three distinct failure modes and show, in a unifying experiment, that Langevin sampling on the GFlowNet-tuned energy is indistinguishable from Langevin sampling on the untuned base model. The pathology survives amortization because it is a property of the training objective of autoregressive language models, which shapes local conditional distributions and never a global energy over sequences. A constrained-generation extension with a sentiment classifier confirms the picture: the constraint gradient carries no reliable steering signal on this energy landscape. The contribution of this thesis is therefore a mechanism, established twice by independent means, for why energy-guided sampling on frozen autoregressive language models fails, together with a positive-control experiment that would falsify it.

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Chapter 1

Introduction

Text generation with large language models has, in the space of a few years, moved from a research curiosity to a technology that underpins a wide range of applications, from writing assistants and conversational agents to summarization, translation, and code synthesis. The dominant paradigm behind these systems is **autoregressive generation**: a model factorizes the probability of a sequence into a product of next-token conditionals and produces text one token at a time, from left to right, sampling or maximizing each token given the tokens already committed [Radford et al., 2019, Brown et al., 2020]. This factorization is what makes modern language models trainable at scale and fast at inference, and it is the reason they work as well as they do.

The same factorization, however, is the source of a persistent and practically important limitation. Because generation proceeds strictly from left to right and each token is fixed once emitted, an autoregressive model has no mechanism for revising an earlier decision in light of a later one. This is not a problem when the objective is simply to produce fluent continuations, but it becomes a problem the moment we want to *control* what the model produces. Many of the properties we care about most, such as the overall sentiment of a passage, its topical focus, its adherence to a factual constraint, or the absence of toxic content, are **global properties** of a completed sequence rather than local properties of any single token. A greedy left-to-right decision about the next token cannot see these properties, because the object that would be evaluated against them, the finished text, does not yet exist. The problem of steering a language model toward such global objectives is the problem of **controllable text generation**, and it is the subject of this thesis.

There are, broadly, two established ways of imposing control, and each carries a cost. The first is to bake the desired behaviour into the parameters of the

model through supervised fine-tuning or reinforcement learning from human feedback [Ouyang et al., 2022]. This is effective, but it is *static*: a model tuned to be helpful and non-toxic embodies exactly those constraints and no others, and a new constraint at inference time requires a new dataset and a new training run. The second is to filter or re-rank many independent generations, discarding those that violate the constraint, which is conceptually simple but computationally wasteful and, for constraints that are rarely satisfied by chance, effectively intractable. What both approaches lack is the ability to introduce a fresh constraint at the moment of generation, cheaply, without touching the underlying model.

1.1 Energy-Based Control and Its Central Assumption

An elegant alternative reframes decoding itself. Rather than generating a sequence token by token, one can define a probability distribution over the space of all sequences and *sample* from it. Following the standard formulation of **energy-based models** [LeCun et al., 2006, Hinton, 2002], the distribution is written in the Boltzmann form

$$(1.1) \quad p(\mathbf{x}) = \frac{1}{Z} \exp(-E(\mathbf{x})),$$

where $E(\mathbf{x})$ is the **energy** of a sequence $\mathbf{x}$, low energy corresponds to a desirable sequence, and Z is a normalizing constant that need never be computed if one only wishes to sample. The attraction of this view for controllable generation is that the energy can be assembled additively from independent terms:

$$(1.2) \quad E(\mathbf{x}) = -\log p_{\text{LM}}(\mathbf{x}) + \lambda C(\mathbf{x}),$$

where the first term, the negative log-likelihood of a frozen pretrained language model, supplies *fluency*, and the second term $C(\mathbf{x})$ is an arbitrary differentiable *constraint* weighted by λ . Because the constraint is simply added on, it can be specified at inference time and swapped freely: a practitioner who wants positive sentiment today and factual consistency tomorrow changes $C(\mathbf{x})$ and leaves the language model untouched. This is the promise of plug-and-play, inference-time controllable generation, and a substantial and growing body of recent work pursues it, notably through gradient-guided decoders such as PPLM [Dathathri et al., 2020], FUDGE [Yang and Klein, 2021], GeDi [Krause et al., 2021], and, most directly related to

this thesis, sampling-based methods such as COLD [Qin et al., 2022] and MuCoLa [Kumar et al., 2022], which draw samples from the energy in (1.2) using continuous relaxations and Langevin-style dynamics.

Every method in this family rests on a single, rarely examined assumption. To sample from the energy in (1.2) with a gradient-guided method, one must be able to compute a gradient of $-\log p_{\text{LM}}(\mathbf{x})$ that points, at least on average, toward better sequences, and one must be able to follow that gradient across the space of possible texts. In other words, the entire framework presupposes that the frozen likelihood of an autoregressive language model, viewed as a function over sequences, defines a *landscape that a sampler can actually navigate*. This assumption is intuitive, and it is almost never tested directly. It is the central object of study in this thesis, and the central finding is that, for standard autoregressive language models on discrete text, the assumption does not hold.

1.2 From Sampling to Amortization

The natural response to a gradient that will not cooperate is to stop using the gradient. If a per-sequence energy can still be *evaluated*, even when it cannot be usefully *differentiated*, then one can treat the language model as a black box that scores finished sequences and learn a separate policy that generates in proportion to that score. This is the idea of **amortized inference**, and its most principled recent realization for sequence generation is the **Generative Flow Network**, or GFlowNet [Bengio et al., 2021, 2023], which trains a generative policy so that the probability of producing a complete sequence is proportional to its reward. Applied to language models, GFlowNets have been proposed as a way to sample diverse, high-reward text without the mode-collapse of reward-maximizing reinforcement learning [Hu et al., 2024]. The second half of this thesis asks whether amortizing the search in this way escapes the difficulties that afflict the gradient-guided samplers, and finds that it does not, because the difficulty lies not in the search procedure but in the energy itself.

1.3 Research Questions and Contributions

The work in this thesis is organized around four research questions.

RQ1. Can the frozen likelihood of an autoregressive language model be sampled

effectively with theoretically faithful Langevin dynamics on the discrete text manifold, and if not, why not?

RQ2. What is the role of the Metropolis–Hastings correction in each of the discrete and continuous samplers, and does making the samplers theoretically correct also make them work?

RQ3. Does amortizing the energy with a GFlowNet, which never differentiates the reward, escape the failures observed for the gradient-guided samplers?

RQ4. Can an additive constraint term steer generation toward a target attribute on this energy landscape, as the plug-and-play framework requires?

In answering these questions the thesis makes the following contributions. It establishes, through a controlled ablation that separates the direction of the gradient from its magnitude, that the frozen autoregressive gradient carries no usable information for discrete sampling, and it replicates this finding across five energy functions. It identifies and measures the *linearization failure* that is the geometric cause of this result, showing that the region over which the sampler’s Taylor surrogate is valid is smaller than the smallest possible discrete step. It measures the *likelihood trap* on the study’s own models, demonstrating that the lowest-energy text is the most degenerate. It provides a precise, measured account of the *Metropolis–Hastings breakdown* in continuous space, attributing the collapse of the acceptance ratio to the proposal term and connecting it to the non-Lipschitz drift induced by nearest-neighbour projection. It documents three mechanistically distinct failure modes of GFlowNet fine-tuning on a frozen-likelihood reward, and it shows in a unifying experiment that Langevin sampling on the amortized energy is indistinguishable from sampling on the untuned base. Finally, it reports a constrained-generation extension that confirms the constraint gradient carries no reliable steering signal, and it proposes a positive-control experiment on a diffusion language model that would falsify the thesis’s central explanation.

1.4 Structure of the Thesis

Chapter 2 introduces the background needed to follow the technical development: energy-based models, Langevin dynamics and the Metropolis–Hastings correction, the discrete and continuous samplers studied here, and GFlowNets. Chapter 3 situates the work within the literature on controllable and energy-based text generation

and reviews the specific difficulties that discreteness and the geometry of language-model embeddings pose for gradient-guided methods. Chapter 4 describes the experimental platform, the masked-token recovery task, the five energy functions, the sampler configurations, the evaluation metrics, and the four diagnostic experiments designed to isolate the mechanisms. Chapter 5 presents the results, beginning with the calibration of the samplers, proceeding through the annealing dynamics, the Metropolis–Hastings breakdown, and the central gradient-fallacy result, and then turning to the GFlowNet experiments and the constrained-generation extension. Chapter 6 draws the mechanisms together into a single account and considers the implications for future work, including the positive-control experiment. Chapter 7 concludes.

Chapter 2

Background Work

This chapter develops the technical background that the remainder of the thesis relies on. It begins with the energy-based view of sequence generation and the way a frozen language model is repurposed as an energy function. It then introduces Langevin dynamics as a gradient-guided sampling method, the Metropolis–Hastings correction that makes such samplers asymptotically exact, and the two concrete samplers, discrete and continuous, that are studied in the experiments. It closes with an account of GFlowNets as an amortized alternative that avoids the gradient altogether. Throughout, the aim is not to survey these tools exhaustively but to expose the specific assumptions each one makes, because it is precisely those assumptions that the experiments in Chapter 5 put under strain.

2.1 Autoregressive Language Models as Energy Functions

An autoregressive language model defines a probability distribution over a sequence of tokens $\mathbf{x} = (x_1, \dots, x_T)$ by the chain rule,

$$(2.1) \quad \log p_{\text{LM}}(\mathbf{x}) = \sum_{t=1}^T \log p_{\text{LM}}(x_t \mid x_{<t}),$$

where each conditional is produced by a neural network, in the models studied here a Transformer [Vaswani et al., 2017], applied to the prefix $x_{<t}$. The model is trained by **teacher forcing**, that is, by maximizing the log-likelihood in (2.1) on a corpus with the true prefix supplied at every step. It is important for what follows to be clear about what this objective does and does not shape. It shapes each conditional

$p_{\text{LM}}(x_t \mid x_{<t})$ to be an accurate predictor of the next token given a correct history. It does not, at any point, ask the joint quantity $\log p_{\text{LM}}(\mathbf{x})$ to behave as a smooth or well-calibrated function of $\mathbf{x}$ when $\mathbf{x}$ is varied off the data manifold. This distinction, between a well-trained local conditional and a well-behaved global energy, is the seed of the central result of the thesis, and we return to it in Chapter 6.

The energy-based reformulation is immediate. Identifying the Boltzmann form of (1.1) with the language model, one sets

$$(2.2) \quad E_{\text{LM}}(\mathbf{x}) = -\log p_{\text{LM}}(\mathbf{x}),$$

so that sequences of high probability are sequences of low energy. Sampling text from the language model is then, formally, sampling from the Boltzmann distribution of this energy. For controllable generation, an additive constraint term $\lambda C(\mathbf{x})$ is appended as in (1.2), where C is typically the negative log-probability that an attribute classifier assigns to the desired label. The task of generation becomes the task of drawing low-energy samples from the combined energy, and the central computational question is how to do so.

2.2 Langevin Dynamics

Suppose the state $\mathbf{s}$ lived in a continuous space $\mathbb{R}^D$ and the energy were differentiable. Then one could sample from $p(\mathbf{s}) \propto \exp(-E(\mathbf{s}))$ using **Langevin dynamics**, a gradient-guided Markov chain Monte Carlo method [Roberts and Tweedie, 1996, Welling and Teh, 2011]. The discretized update is

$$(2.3) \quad \mathbf{s}_{t+1} = \mathbf{s}_t + \frac{\epsilon_t}{2} \nabla_{\mathbf{s}} \log p(\mathbf{s}_t) + \sqrt{\epsilon_t} \boldsymbol{\xi}_t, \quad \boldsymbol{\xi}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

where ϵ_t is a step size at iteration t . The update has two parts, and the interplay between them is what distinguishes sampling from optimization. The first part, the **drift**, is a step in the direction of the gradient of the log-density; on its own it is gradient ascent and it drives the state toward a single mode. The second part, the **diffusion**, injects Gaussian noise scaled by $\sqrt{\epsilon_t}$, and it is what allows the chain to escape local optima and explore. If the step size is annealed toward zero on an appropriate schedule, the distribution of $\mathbf{s}_t$ converges to the target $p(\mathbf{s})$ rather than collapsing onto its mode [Welling and Teh, 2011]. A useful intuition is that of a marble on an uneven surface in a room that is being shaken: while the shaking is vigorous the marble explores the whole surface, and as the shaking is gradually

stilled, the resting place of the marble is a fair sample from the landscape rather than merely its lowest point.

The convergence guarantee, however, rests on a regularity condition that will matter a great deal in Chapter 5. For the Metropolis-adjusted variant discussed below to be valid, the drift term must be Lipschitz continuous [Roberts and Tweedie, 1996]; loosely, the gradient must not change too abruptly from one point to a nearby one. When the energy landscape is smooth this is unproblematic. When, as in the continuous sampler of Section 2.4, the energy is defined through a discontinuous projection, the condition fails, and the failure is not benign.

2.3 The Metropolis–Hastings Correction

The update in (2.3) is a discretization, and any finite step size introduces a bias: the chain does not sample exactly from $p(\mathbf{s})$ but from a perturbed distribution. The **Metropolis–Hastings** (MH) correction removes this bias [Metropolis et al., 1953, Hastings, 1970]. Having proposed a move from $\mathbf{s}$ to $\mathbf{s}'$ according to a proposal density $q(\mathbf{s}' | \mathbf{s})$, one accepts the move with probability

$$(2.4) \quad \alpha(\mathbf{s}' | \mathbf{s}) = \min \left\{ 1, \frac{p(\mathbf{s}') q(\mathbf{s} | \mathbf{s}')}{p(\mathbf{s}) q(\mathbf{s}' | \mathbf{s})} \right\},$$

and otherwise remains at $\mathbf{s}$. The acceptance ratio factors into two parts. The **target ratio** $p(\mathbf{s}')/p(\mathbf{s})$ measures whether the proposed state is more probable under the distribution being sampled. The **proposal ratio** $q(\mathbf{s} | \mathbf{s}')/q(\mathbf{s}' | \mathbf{s})$ measures whether the move is as easy to reverse as it was to make, and it is what enforces *detailed balance*, the condition that guarantees the chain has p as its stationary distribution. When Langevin dynamics is equipped with this correction the method is known as the Metropolis-adjusted Langevin algorithm, or MALA.

Two points about the correction are worth stating plainly, because both recur in the experiments. First, many energy-based decoders in the literature omit the correction entirely; without it, the procedure is a biased optimizer dressed as a sampler, and the distinction is not merely academic. Second, the proposal ratio is subtle to compute correctly for a Langevin proposal, because the reverse proposal density $q(\mathbf{s} | \mathbf{s}')$ must be evaluated using the drift computed *at the proposed point* $\mathbf{s}'$. If the drift at $\mathbf{s}'$ is wildly different from the drift at $\mathbf{s}$, as it is when the landscape is discontinuous, the reverse proposal can place the original state $\mathbf{s}$ deep in the tail of a Gaussian, driving the proposal ratio, and with it the acceptance probability, to

zero. This is the mechanism that paralyzes the continuous sampler, and Chapter 5 measures it directly.

2.4 Sampling in a Discrete Space: DLS and CLS

The difficulty that motivates the entire thesis is that text is not continuous. The state is a sequence of discrete tokens drawn from a finite vocabulary V , and there is no state *between* two tokens. Langevin dynamics, which assumes a continuous differentiable space, cannot be applied to such a state without modification. Two strategies for adapting it are studied here, and they differ in where they confront the discreteness.

The **Discrete Langevin Sampler** (DLS), following the discrete gradient-based samplers of Grathwohl et al. [2021] and Zhang et al. [2022], never leaves token space. It uses the gradient of the log-likelihood with respect to the input embedding of a masked position only as a *score*, to construct a categorical proposal distribution over which token to place there next. Concretely, if $\mathbf{e}(x_i)$ is the embedding of the current token at a masked position i and $\mathbf{g} = \nabla_{\mathbf{e}_i} \log p_{\text{LM}}(\mathbf{x})$ is the gradient at that position, the proposal assigns to a candidate token v a logit that combines a term penalizing the embedding distance $\|\mathbf{e}(v) - \mathbf{e}(x_i)\|$ with a term proportional to $\frac{1}{2}\mathbf{g}^\top(\mathbf{e}(v) - \mathbf{e}(x_i))$. The second of these is a **first-order Taylor surrogate** for the change in log-likelihood that would result from the substitution, and its validity is the subject of the central diagnostic of the thesis.

The **Continuous Langevin Sampler** (CLS), following the continuous-relaxation approach of methods such as MuCoLa [Kumar et al., 2022], takes the opposite route. It relaxes the state into the continuous embedding space $\mathbb{R}^D$, runs genuine Langevin dynamics there according to (2.3), and projects the continuous state back onto the nearest token embedding when a discrete sequence is required. The energy it navigates is therefore defined through a **nearest-neighbour projection** into the vocabulary. This construction has an immediate geometric consequence, developed at length in Chapter 5: the embedding space is partitioned into **Voronoi cells**, one per token, within which the projected token, and hence the energy, is constant, and across whose boundaries the energy changes discontinuously. The landscape the continuous sampler must traverse is thus not a smooth surface but a collection of flat plateaus separated by cliffs, and it is on exactly such a landscape that the Lipschitz condition of Section 2.3 fails.

2.5 Amortized Inference with GFlowNets

Both samplers above require the gradient of the energy. An entirely different strategy is to give up on the gradient and instead *learn* to sample. A **Generative Flow Network** [Bengio et al., 2021, 2023] treats generation as a sequential process of constructing an object step by step, and trains a policy so that the probability of terminating at a complete object is proportional to a non-negative reward $R(\mathbf{x})$. The name reflects a useful metaphor: one imagines a fixed quantity of flow entering at an initial state and being divided at each branching point by the policy, so that the amount of flow arriving at each terminal object is proportional to its reward. Generation then amounts to following the flow, and because the policy allocates flow in proportion to reward rather than concentrating it on the single highest-reward object, the resulting samples are *diverse*, which is the property that distinguishes a GFlowNet from a reward-maximizing reinforcement-learning agent that would converge on one mode.

For text, the object is a sequence and the construction proceeds left to right, so the generative structure is a tree: there is exactly one path to any given sequence. The training objectives used in practice, such as trajectory balance and its sub-trajectory variants [Malkin et al., 2022, Madan et al., 2023], enforce a flow-consistency condition along trajectories. The crucial feature of the whole approach for this thesis is that the reward $R(\mathbf{x})$ is only ever *evaluated*, never differentiated: the discontinuity and non-smoothness of the underlying energy landscape, which cripple the gradient-guided samplers, are simply invisible to a GFlowNet. If the difficulty with energy-guided sampling were purely a property of the gradient, amortization ought to escape it. Whether it does is the question of Section 5.8. Following Hu et al. [2024], the GFlowNet is realized as a parameter-efficient adapter [Hu et al., 2022] on the same frozen base model that supplies the energy, so that the two halves of the study share a model and the comparison between them is not confounded.

Chapter 3

Related Work

The work in this thesis sits at the intersection of three lines of research: controllable text generation, energy-based and sampling-based decoding, and the study of the geometry of language-model representations. This chapter reviews each in turn, with an emphasis on the specific claims and assumptions that the experiments later test, and it closes by drawing out the gap in the literature that the thesis addresses.

3.1 Controllable Text Generation

Approaches to controllable generation can be organized by how much they change the underlying model. At one extreme, **fine-tuning methods** adjust the model’s parameters to internalize the desired behaviour, either by supervised training on curated data or, more commonly today, by reinforcement learning from human feedback [Ouyang et al., 2022]. These methods are effective and now standard, but the control they provide is fixed at training time; introducing a new attribute requires new data and further training. **Prompting and conditioning methods** such as CTRL [Keskar et al., 2019] steer generation with control codes prepended to the input, which is inexpensive but limited to attributes anticipated during training.

At the other extreme are **decoding-time methods**, which leave the model frozen and intervene only during generation. These are the methods most relevant to this thesis, because they are the ones that promise inference-time flexibility. PPLM [Dathathri et al., 2020] perturbs the hidden states of a frozen model in the direction of an attribute classifier’s gradient at each step. FUDGE [Yang and Klein, 2021] and GeDi [Krause et al., 2021] reweight the next-token distribution using a discriminator that predicts whether a prefix will lead to the desired attribute. These methods remain autoregressive: they modify the local next-token decision but retain

the strictly left-to-right structure and therefore its inability to revise. The sampling-based methods discussed next are an attempt to escape that structure entirely.

3.2 Energy-Based and Sampling-Based Decoding

A distinct and influential line of work casts decoding as sampling from an energy defined over whole sequences, as in (1.2), and thereby allows global constraints to be imposed on the completed text rather than token by token. COLD [Qin et al., 2022] performs energy-based constrained decoding by running Langevin dynamics on a relaxed representation of the sequence, combining a fluency energy from the language model with task-specific constraint energies. MuCoLa [Kumar et al., 2022] samples in the continuous embedding space and projects back to tokens, and it is the closest published antecedent of the continuous sampler studied here. Related efforts include gradient-based inference for infilling and lexically constrained generation [Sha, 2020] and a range of discrete samplers adapted from the energy-based-model literature, such as the gradient-informed discrete proposals of Grathwohl et al. [2021] and the discrete Langevin proposals of Zhang et al. [2022] on which the discrete sampler here is based.

A recurring and rarely foregrounded feature of this literature is that the Metropolis–Hastings correction is frequently omitted. Methods are often described as sampling from the energy while in practice running a biased, noise-augmented optimizer, a distinction discussed in Section 2.3. Where the correction is included, the difficulty of computing the reverse proposal on a landscape defined by projection is seldom examined. One contribution of this thesis is to take the correction seriously, to implement it faithfully for both samplers, and to measure what happens when it is applied to a landscape that violates the smoothness it presupposes.

3.3 Amortized Sampling and GFlowNets

Generative Flow Networks were introduced by Bengio et al. [2021] as a method for sampling discrete compositional objects in proportion to a reward, with the explicit goal of producing diverse high-reward samples rather than the single mode that reward-maximizing reinforcement learning tends to find. The framework was formalized and extended in subsequent work [Bengio et al., 2023], and a family of training objectives was developed, including trajectory balance [Malkin et al., 2022] and sub-trajectory balance [Madan et al., 2023], which improve credit assignment

along generative trajectories. Hu et al. [2024] applied GFlowNets to language models, framing tasks such as infilling and reasoning as sampling from an intractable posterior defined by the frozen model, and reported that GFlowNet fine-tuning yields diverse samples that transfer better than those from reward-maximizing baselines. The GFlowNet component of this thesis builds directly on their formulation and codebase, using the same modified sub-trajectory objective and the same parameter-efficient adaptation of the base model [Hu et al., 2022]. The question posed here is narrower and more critical than theirs: not whether amortized sampling is useful in general, but whether it escapes the specific pathology of the frozen-likelihood energy that defeats the gradient-guided samplers.

3.4 The Geometry of Language-Model Representations

A final line of work concerns the geometry of the embedding spaces in which the continuous sampler operates, and it supplies part of the explanation for the results in Chapter 5. A well-documented property of contextual and static embeddings from Transformer language models is **anisotropy**: embeddings do not spread uniformly through the representation space but occupy a narrow cone, so that randomly chosen tokens have high average cosine similarity [Gao et al., 2019, Ethayarajh, 2019]. Anisotropy has direct consequences for any method that treats Euclidean or cosine distance in embedding space as a meaningful measure of semantic similarity, because it compresses the dynamic range of those distances. This thesis measures the anisotropy of the two models it studies and shows that it renders Euclidean distance an unreliable evaluation metric, which is one reason the experiments rely on a divergence-based measure of contextual fit instead.

A second and complementary body of work concerns the failure of high-probability text to be high-quality text. Holtzman et al. [2020] showed that the most probable sequences under a neural language model are often degenerate, repetitive, and unnatural, and that maximization-based decoding such as beam search produces markedly worse text than sampling. This observation, which the present thesis reproduces on its own models and terms the *likelihood trap*, is central to the interpretation of the results: it means that any procedure which succeeds in concentrating probability mass on the lowest-energy regions of the frozen-likelihood energy is thereby concentrating it on degenerate text.

3.5 The Gap Addressed by This Thesis

The literature reviewed above establishes energy-based sampling as an attractive route to controllable generation and provides both the samplers and the amortized alternative that this thesis studies. What it does not provide is a direct, controlled test of the assumption on which the whole enterprise rests, namely that the gradient of a frozen autoregressive likelihood is a usable search direction on the discrete text manifold, together with a mechanistic account of what happens when that assumption fails. Prior negative observations tend to be reported in passing, as brittleness or as tuning difficulty, rather than isolated and explained. This thesis addresses that gap. It designs an ablation that separates the direction of the gradient from its magnitude and thereby tests the assumption cleanly; it constructs diagnostic experiments that identify the geometric, statistical, and analytic mechanisms behind the result; and it shows that the same mechanism survives amortization, thereby locating the cause in the training objective of the language model rather than in any particular sampling algorithm.

Chapter 4

Methodology

This chapter describes the experimental platform in the detail needed to reproduce the study. It sets out the diagnostic task and why it was chosen, the five energy functions, the two samplers and the factorial space of configurations in which they were run, the evaluation metrics and the reasoning behind them, the GFlowNet training setup, the constrained-generation extension, and finally the four diagnostic experiments that were designed specifically to isolate the mechanisms behind the main results.

4.1 Task: Masked Token Recovery

A sampler can only be studied if there is a way to tell whether it is being guided toward good sequences or merely wandering. This requires a task with a known target. We therefore adopt **masked token recovery**, also called infilling, as the diagnostic bench. A sequence is drawn from the corpus, one or more of its tokens are replaced with random tokens, and the sampler is asked to restore a coherent sequence by resampling only the corrupted positions while the remainder is held fixed. Because the original sequence is known, the recovery can be evaluated, and because the corruption and the context are controlled, the behaviour of the sampler can be attributed to the method rather than to the difficulty of an open-ended generation.

An important subtlety shapes the choice of evaluation metric and is worth stating at the outset. The token a sampler recovers need not be the original token in order to be correct. If the corrupted word *roommate* is recovered as *dorm*, the sequence is perfectly coherent even though it does not match the reference. An exact-match criterion would wrongly score this as a failure. For this reason the primary evaluation

is based not on token identity but on how well the recovered sequence fits its context under the model’s own predictive distribution, as described in Section 4.4. The corpus is **ROCStories** [Mostafazadeh et al., 2016], a collection of short five-sentence everyday narratives. It is chosen for three reasons that matter for a controlled diagnostic: its sentences are short and syntactically simple, so a single corrupted token has a well-defined and locally checkable correct recovery; its domain is narrow and consistent, so the model’s uncertainty reflects the sampler’s behaviour rather than the open-endedness of the text; and it is the corpus used by the amortized-inference reference implementation of Hu et al. [2024], so the GFlowNet half of the study operates on the same distribution as the work it builds on. Masked recovery, rather than open-ended generation, is chosen as the diagnostic task precisely because it supplies a known target: a sampler can only be judged to be guiding well or poorly if there is a ground truth against which its output can be scored, and open-ended generation provides none.

4.2 Energy Functions

The study evaluates five energy functions. The reference energy is a **GPT-2 Large** model [Radford et al., 2019], of 774 million parameters and embedding dimension $D = 1280$, supervised fine-tuned on ROCStories; this same fine-tuned model is used throughout as the base for the amortized experiments, so that the Langevin and GFlowNet halves of the study share weights and are not confounded by a difference in the underlying model. Three **GFlowNet-tuned variants** of this base model are included, obtained as described in Section 4.5: two trained with the unnormalized length setting, at 500 and 2000 optimization steps respectively, and one trained with the length-normalized setting at 500 steps. Finally, a **Llama-3 8B** model [Dubey et al., 2024], with embedding dimension $D = 4096$, serves as a cross-architecture control. Because Llama-3 uses a different tokenizer and a different embedding geometry, and consequently a different step-size scale, it is never placed on a shared numerical axis with the GPT-2 family; it is used to establish that qualitative findings hold across architectures. The four GPT-2-based models share the tokenizer, the corruption seeds, and the step-size schedule, and their numbers are therefore directly comparable without qualification.

4.3 Samplers and Configurations

The two samplers of Section 2.4, the Discrete Langevin Sampler (DLS) and the Continuous Langevin Sampler (CLS), are each run under a factorial combination of design choices. The two are studied together, rather than either alone, because they represent the two opposite strategies for confronting the discreteness of text and therefore isolate different failure modes: the DLS, following Grathwohl et al. [2021] and Zhang et al. [2022], keeps the state in token space and uses the gradient only to build a proposal, which tests whether the gradient is an informative ranking signal; the CLS, following Kumar et al. [2022], relaxes into continuous space and runs genuine Langevin dynamics, which tests whether continuous Markov chain Monte Carlo is compatible with a projected discrete energy. A finding that holds for both is a finding about the energy rather than about one sampler’s idiosyncrasies, which is why both are needed. The **proposal method** is one of three: *policy*, which uses the true gradient of the energy; *grad-norm-preserved random*, which replaces the gradient with a random vector rescaled to the exact norm of the true gradient; and *random*, which uses a random vector of random magnitude. The comparison between these three is the central ablation of the thesis, isolating respectively the full gradient, the effect of its direction, and a pure baseline. The **Metropolis–Hastings correction** is either enabled or disabled. **Gradient normalization**, which rescales the gradient to unit norm for numerical stability, is either enabled or disabled; this choice interacts with the proposal-method ablation in a way that turns out to be critical, as explained in Section 5.4. The **annealing schedule** runs for either 50 or 100 steps, with the step size decreasing linearly; for the GPT-2 family the schedule runs from $\epsilon_{\text{start}} = 10.5$ to $\epsilon_{\text{end}} = 0.1$, a scale established by the calibration described in Section 5.1. An *oracle* variant, in which the ideal step size at each iteration is selected using knowledge of the ground-truth token, is included as an upper-bound reference; its purpose is to rule out the possibility that the samplers fail merely for want of a well-tuned schedule, since if even an oracle-tuned schedule does not let the gradient outperform a random direction, the failure cannot be one of calibration. In total, across the five energy functions, the study comprises 145 sampling configurations, each evaluated on 200 sequences.

4.4 Evaluation Metrics

Three quantities are recorded. The first is the **Euclidean distance** in embedding space between the recovered token and the ground-truth token. This is the most

obvious measure of recovery, but it proves to be unreliable, for a reason established quantitatively in Section 5.7: because language-model embeddings are anisotropic, occupying a narrow cone, Euclidean distance compresses the distinction between good and bad tokens, and a token can be geometrically close to the target while being grammatically inadmissible. It is reported for completeness and as an object of study in its own right, not relied upon as the primary criterion.

The primary criterion is the **KL divergence** between the model’s predictive distribution at the recovered position and a reference distribution, which measures how well the recovered token fits its context, that is, its *contextual coherence*. A recovered sequence that the model finds locally natural has low divergence regardless of whether it matches the reference token, which is the property required of a metric on this task. We note one limitation of the divergence measure, which is that its magnitude depends on the position of the corrupted token within the sequence: corrupting an early token perturbs the conditional probability of a long suffix and yields a large divergence, whereas corrupting a final token yields almost none. For this reason no model is ranked on small differences in KL divergence; the central results are argued from the *absence* of a gap between conditions rather than from a small measured gap, which is robust to this limitation in a way that a small measured difference would not be. The KL divergence [Kullback and Leibler, 1951] is chosen as the primary metric over exact-match accuracy because, as noted above, a recovered token that differs from the reference may still be perfectly coherent, and a metric that penalized such recoveries would confound the quality of the sampler with the arbitrariness of the reference; a divergence between predictive distributions measures contextual fit directly and is insensitive to which particular admissible token was chosen. Finally, for the generation-quality analyses, we record **repetition rate**, the fraction of repeated n -grams, and **distinct- n** , which are the standard measures of the degeneration studied by Holtzman et al. [2020] and are therefore the appropriate instruments for detecting the likelihood trap; and for the constrained-generation extension we record classifier accuracy and **BERTScore** [Zhang et al., 2020], the latter chosen because it measures semantic similarity through contextual embeddings rather than surface overlap and so does not penalize valid paraphrases of the reference.

4.5 GFlowNet Training

The GFlowNet is chosen as the amortized alternative, over reward-maximizing reinforcement learning, for a reason specific to the question of this thesis: a reward-maximizing agent would converge on the single highest-reward sequence and so could not distinguish a sampler that explores the energy from one that merely finds its mode, whereas a GFlowNet [Bengio et al., 2021, 2023] is trained to sample in proportion to reward and therefore tests amortized *sampling* rather than amortized optimization. Crucially for the argument, it only ever evaluates the reward and never differentiates it, so it is blind to exactly the gradient pathology that afflicts the Langevin samplers, which makes it the ideal instrument for asking whether that pathology can be sidestepped. The GFlowNet-tuned variants are produced following the amortized-inference formulation of Hu et al. [2024], using their modified sub-trajectory-balance objective [Malkin et al., 2022, Madan et al., 2023]. The policy is a low-rank adapter [Hu et al., 2022] on the frozen GPT-2 Large base, and the reward is the log-likelihood assigned by that frozen base to the completed sequence, so that the GFlowNet is trained to sample in proportion to the model’s own likelihood. The task is a story-infilling formulation in which, given a preceding context X and a known ending Y , the policy generates a connecting span Z , and the reward scores the concatenation. A single hyperparameter, denoted `len_beta`, controls length normalization of the reward: at `len_beta` = 0 the reward is the raw unnormalized log-likelihood, and at `len_beta` = 1 it is fully normalized by sequence length. The consequences of this single knob are one of the central findings of the GFlowNet study, developed in Section 5.8.

4.6 Constrained Generation Extension

To test the plug-and-play claim directly (RQ4), a constrained-generation extension augments the energy with a sentiment constraint. A sentiment classification head is trained on the representations of the GPT-2 Large base and used to construct the constraint term $C(\mathbf{x})$ in (1.2). Generation is then run under five constraint modes that isolate the contribution of the constraint gradient: *lm_only*, which uses only the fluency energy; *cons_only*, which uses only the constraint; *full*, which combines them with weights $\beta_{\text{LM}} = 0.8$ and $\beta_{\text{C}} = 0.2$; *random*, which replaces the combined gradient with noise; and *cons_random*, which replaces only the constraint gradient with noise while retaining the fluency gradient. The steering effect is measured as the change in the fraction of generations classified as the target sentiment relative

to the unconstrained baseline. Both the discrete sampler in the formulation of this thesis and a MuCoLa-style continuous baseline are evaluated, on both infilling and continuation.

4.7 Diagnostic Experiments

The configurations above establish *that* the methods fail. To establish *why*, four additional diagnostic experiments were designed, each targeting one candidate mechanism. They do not sample; they measure properties of the energy landscape and of the gradient directly.

The **linearization experiment** tests whether the first-order Taylor surrogate that the discrete proposal uses to rank candidate tokens actually predicts the true change in sequence energy. For each of 200 sequences and a masked position, the exact gradient at that position is computed, and for a stratified set of 2000 candidate tokens the surrogate predicted change $\hat{\Delta}(v) = \mathbf{g}^\top(\mathbf{e}(v) - \mathbf{e}(x_i))$ is compared against the true change $\Delta(v) = \log p(\mathbf{x}_{i \rightarrow v}) - \log p(\mathbf{x})$ obtained by an exact forward pass, and the embedding distance of each candidate is recorded so that the correlation can be examined as a function of distance. The true change is further decomposed into a *self* term, the change in the log-probability of the candidate given its own left context, and a *future* term, the change in the log-probability of the suffix, because the gradient with respect to the input embedding is structurally able to see only the latter.

The **acceptance experiment** tests the Metropolis–Hastings breakdown in the continuous sampler by logging, for every proposal, whether it crossed a Voronoi cell boundary, whether it was accepted, and the decomposition of the acceptance ratio into its target and proposal terms. The **trajectory experiment** logs the raw state of the sampler at every step for a small number of sequences, so that the path can be projected into two dimensions with both PCA and t-SNE, and it records the distance from the continuous state to the nearest token embedding, quantifying how far the sampler drifts from the token manifold. The **likelihood-trap experiment** decodes with several strategies, from greedy and beam search to ancestral sampling, and measures the relationship between the per-token likelihood of the generated text and its repetition rate, establishing on the study’s own models whether the lowest-energy text is in fact the worst text. A final **anisotropy analysis** measures the distribution of inter-token distances and related geometric quantities in the two embedding spaces. All experiments use a resumable file-queue system with per-job

VRAM gating, and all sampling code was verified against a reference implementation with a bitwise-equivalence test suite before the runs reported here were launched.

Chapter 5

Results

This chapter presents the experimental findings. It proceeds in the order in which the study developed, because that order is also the logical one. It begins with the calibration of the samplers and the discovery that the step size does not transfer between models, which already reveals the influence of embedding geometry. It then examines the annealing dynamics of the discrete sampler and shows that its apparent convergence is an artifact of cooling. It turns next to the Metropolis–Hastings correction and shows that the same correction which helps the discrete sampler destroys the continuous one, and it measures precisely why. The central section establishes the gradient fallacy: that following the true gradient is indistinguishable from following a random direction, on all five energy functions. Three diagnostic sections then supply the mechanism, through the linearization radius, the likelihood trap, and the anisotropy of the embedding space. The chapter closes with the GFlowNet experiments, which show that amortization does not escape the pathology, and the constrained-generation extension, which confirms that the constraint gradient carries no reliable steering signal. Throughout, each reported number is annotated in the source of this document with the results file from which it was drawn, so that every quantitative claim is traceable to the experiment that produced it.

5.1 Step-Size Calibration and Embedding Geometry

The first attempts to run the samplers on GPT-2 Large, using a step-size schedule that had worked for Llama-3, failed completely: the entropy of the proposal collapsed to zero within a few steps and no token was ever changed. The sampler

was not broken but trapped, and the reason is geometric. The anisotropy analysis, reported in full in Section 5.7, measures the mean distance between neighbouring token embeddings as approximately 1.82 in GPT-2 Large and approximately 0.59 in Llama-3, a difference of roughly a factor of three, with the mean pairwise distance differing similarly (2.77 against 0.84). A step size calibrated for the tighter geometry of Llama-3 is simply too small to carry the continuous state across a Voronoi cell boundary in the more dispersed geometry of GPT-2, so the projected token never changes. Recovering the correct scale by an oracle sweep yielded a schedule running from $\epsilon_{\text{start}} = 10.5$ to $\epsilon_{\text{end}} = 0.1$ for the GPT-2 family, which is used throughout.

The oracle configuration, in which the ideal step size at each iteration is selected using knowledge of the ground-truth token, is included precisely for this calibration purpose: it establishes an upper bound on what any step-size schedule could achieve and thereby separates failures of calibration from failures of method. Even under the oracle, and as reported in the sections that follow, the policy gradient does not outperform a random direction, which is what licenses the conclusion that the difficulty is not one of tuning.

The immediate practical lesson is that there is no universal step-size schedule for these methods; the correct scale is a property of the specific model’s embedding geometry and must be rediscovered per model. This already qualifies the plug-and-play framing, since a method that must be recalibrated to each model’s geometry before it will move at all is not, in the strong sense, model-agnostic. This finding answers the first half of RQ1: theoretically faithful Langevin dynamics cannot be applied off the shelf, because the step size that makes the sampler move at all is model-specific.

5.2 Annealing Dynamics and the Quenching Effect

Run without the Metropolis–Hastings correction, the discrete sampler exhibits a characteristic pattern: its metrics remain noisy and unimproving for the majority of the schedule and then drop abruptly near the end. The natural interpretation is that the sampler has finally converged. A controlled test refutes this interpretation. If the drop marked genuine convergence, it would occur when the model had found a good sequence; if it were instead an artifact of the annealing schedule, it would track the schedule. Extending the schedule from 50 to 100 steps moved the drop from around step 40 to around step 85: it followed the schedule rather than staying fixed. The drop is therefore not convergence but a **quenching effect**. Early in the

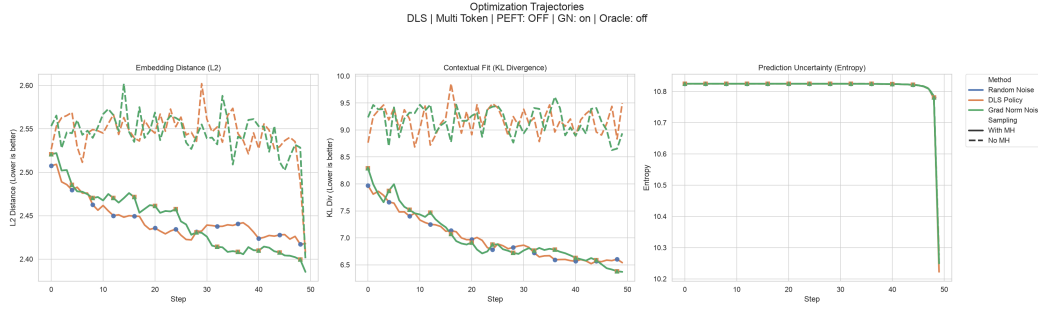


Figure 5.1: Discrete Langevin Sampler trajectories on GPT-2 Large over a 50-step schedule. The metrics remain unimproving for most of the run and drop abruptly near the end as the noise schedule approaches zero, the signature of quenching rather than convergence.

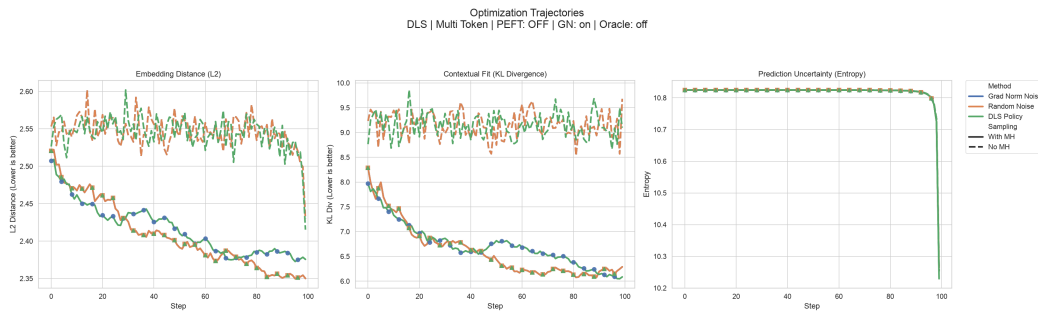


Figure 5.2: The same sampler over a 100-step schedule. The abrupt drop has moved from around step 40 to around step 85, tracking the annealing schedule rather than remaining fixed. This confirms that the drop is an artifact of cooling and not evidence of model convergence.

schedule the step size is large and the injected noise dominates, so the sampler bounces stochastically across the vocabulary; as the schedule anneals toward zero the noise vanishes and the update degenerates into deterministic gradient descent, which collapses greedily into the nearest local optimum. The abrupt late-schedule drop is the signature of this collapse, and Figures 5.1 and 5.2 show the effect at the two schedule lengths. The name is borrowed from metallurgy, where quenching denotes rapid cooling that freezes a material into whatever configuration it happened to occupy, and the analogy is exact: removing the noise freezes the sampler into its current local optimum rather than allowing it to continue exploring.

With the Metropolis–Hastings correction enabled, this pattern does not occur: the chain converges smoothly and early and remains stable. In the discrete sampler, therefore, the theoretically motivated correction is also the empirically better choice. This is worth emphasizing precisely because it is the exception; it is the one place in the study where making the sampler correct also makes it work. The next section shows that the same correction has the opposite effect on the continuous sampler.

5.3 The Metropolis–Hastings Breakdown in Continuous Space

The single most striking result of the sampler study concerns what happens when the Metropolis–Hastings correction is applied to the continuous sampler. Whereas in the discrete sampler the correction acts as a beneficial gatekeeper, in the continuous sampler it is catastrophic: the sampler flatlines, rejecting essentially every proposal that would change anything. The trace experiment quantifies this precisely. Over 10,000 logged proposals from the continuous sampler with gradient normalization disabled and the correction enabled, the overall acceptance rate is 2.6%. Decomposed by whether the proposal crossed a Voronoi cell boundary, the acceptance rate for proposals that *stayed* within a cell, and therefore changed no token, is 0.03%, while the acceptance rate for proposals that *crossed* a boundary is 3.7%. In other words, almost every accepted move is one that leaves the projected token unchanged: the set of moves the correction accepts and the set of moves that do anything are very nearly disjoint. For comparison, the discrete sampler under the correction accepts within-cell moves with probability 1.0 and crossing moves at a healthy rate, which is why it works.

The decomposition of the acceptance ratio explains why, and it locates the cause unambiguously in the proposal term rather than the target term. For proposals that crossed a boundary, the mean *target* log-ratio is +4.60: these moves, on average, *improve* the sequence probability, so the distribution being sampled would welcome them. Yet the mean *proposal* log-ratio for the same moves is -1325 . The reverse proposal assigns effectively zero density to the return move, and this alone drives the acceptance probability to zero. Figure 5.3 shows the two distributions: the target term is centred near zero and frequently positive, while the proposal term for boundary-crossing moves is enormous and negative.

This is the measured form of the Lipschitz argument of Section 2.3, which follows the convergence theory of Roberts and Tweedie [1996]: the Metropolis-adjusted Langevin algorithm requires a Lipschitz-continuous drift, and the energy of the continuous sampler does not have one. Because the energy is defined through a nearest-neighbour projection, its drift is discontinuous at every cell boundary. When a proposal crosses a boundary, the reverse proposal mean is computed from the drift on the far side, which points somewhere unrelated, so the original state falls deep in the tail of the reverse Gaussian and the reverse density vanishes. The correction is doing exactly what it is designed to do; it is the landscape that violates the assumption

Decomposing the acceptance ratio: the target term is fine, the proposal term is fatal

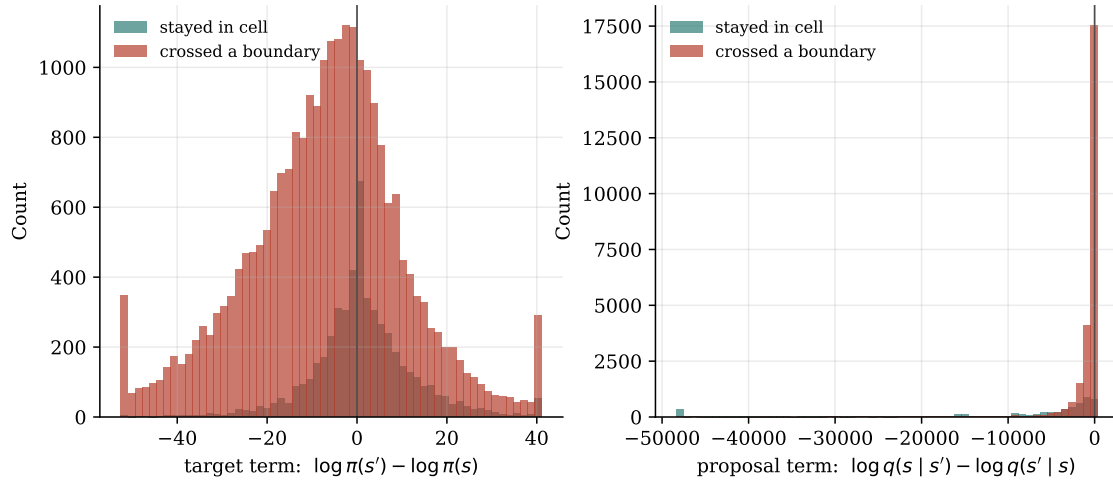


Figure 5.3: Decomposition of the Metropolis–Hastings acceptance ratio in the continuous sampler, for proposals that stay within a Voronoi cell against those that cross a boundary. The target term (left) is well-behaved and often favourable for boundary-crossing moves; the proposal term (right) collapses to large negative values for exactly those moves. The rejection is driven by the proposal term, not the target term.

the correction depends on. The consequence is that the continuous sampler only appears to function when the correction is omitted, that is, when it is run as a biased optimizer. Making it a correct sampler paralyzes it. The same behaviour is observed on both GPT-2 Large and Llama-3, so it is a property of the problem class rather than of a single model, and it is best summarized as a fundamental incompatibility between continuous Markov chain Monte Carlo and a discrete text representation accessed by projection. This answers RQ2 for the continuous sampler: making it theoretically correct does not make it work; it does the opposite.

5.4 The Gradient Fallacy

The results so far concern how the samplers move. The central result concerns whether the gradient they follow carries any useful information at all. The ablation of Section 4.3 compares three proposal methods: the true gradient (*policy*), a random direction rescaled to the gradient’s norm (*grad-norm-preserved*), and a fully random vector (*random*). This three-way design is deliberate and is the methodological core of the thesis, because it separates the two things a gradient can contribute. If the gradient’s direction is informative, the policy method should outperform the norm-preserved random method, which differs from it only in direction; if the magnitude is informative, the norm-preserved method should outperform the fully random

Energy function	Policy	Grad-norm-preserved	Difference
GPT-2 Large (SFT)	6.474	6.430	+0.044
GFlowNet, <code>len_beta=0, 500</code>	6.009	6.001	+0.008
GFlowNet, <code>len_beta=0, 2000</code>	6.698	6.552	+0.146
GFlowNet, <code>len_beta=1, 500</code>	6.008	5.338	+0.670

Table 5.1: Final KL divergence for the true gradient (policy) against a norm-preserved random direction, with gradient normalization disabled and the Metropolis–Hastings correction enabled, on the discrete sampler at 50 steps. Lower is better. Source: `final_kl_by_model.csv`. The differences are small and inconsistent in sign, indicating that the direction of the gradient contributes no reliable improvement over a random direction of equal magnitude.

method, which differs from it only in magnitude.

Before the comparison can be made, a confound must be removed, and identifying it is part of the contribution. Gradient normalization, applied for numerical stability, rescales every proposal direction to unit norm. Under normalization, the norm-preserved random method and the fully random method become the same object, a random unit vector, so an observed equality between the policy and the random methods is consistent with two different explanations: either the direction of the gradient is uninformative, or the normalization has discarded a signal that lived in the magnitude. The normalized ablation cannot distinguish these. The extended ablation therefore repeats the comparison *with gradient normalization disabled*, so that the policy and norm-preserved methods carry the gradient’s full unnormalized magnitude and differ only in direction. Any gap between them is then attributable to direction alone. This is the reason the study comprises 145 configurations rather than fewer, and it is the single design decision that turns a suggestive observation into a claim that survives scrutiny.

With the confound removed, the result is unambiguous. Table 5.1 reports the final KL divergence for the policy and norm-preserved methods with normalization disabled and the correction enabled, together with their difference, across four energy functions. The differences are small and inconsistent in sign. In every case, following the exact gradient is statistically indistinguishable from following a random direction of the same magnitude. The direction of the frozen autoregressive gradient carries no usable information for discrete sampling. Because the norm-preserved and fully random methods are also indistinguishable, the magnitude carries none either.

There is a single statistically significant contrast in the entire matrix, and it points the wrong way for gradient guidance. On Llama-3, with the correction disabled, the true gradient produces a *higher* final KL divergence than a random di-

rection; the gradient is there not merely uninformative but actively misleading, a possibility we explain mechanistically in the next section, since a gradient that points toward the minimizer of a local linearization of a globally discontinuous landscape may point somewhere systematically unhelpful. This answers the second half of RQ1: the gradient is not a usable search direction, and the ablation locates the failure in the direction of the gradient itself rather than in any incidental choice of scale or correction.

5.5 The Linearization Radius

Why does the gradient carry no information? The linearization experiment answers this directly by testing the surrogate the discrete proposal actually uses, following the discrete-sampler construction of Zhang et al. [2022]. Recall that the proposal ranks a candidate token v using the first-order term $\hat{\Delta}(v) = \mathbf{g}^\top(\mathbf{e}(v) - \mathbf{e}(x_i))$, a linear approximation to the true change in log-likelihood. Figure 5.4 plots this surrogate against the true change $\Delta(v)$ over all candidate–sequence pairs for GPT-2 Large. There is no relationship: the Spearman correlation between the surrogate and the truth is -0.011 , which is to say zero. Ranking candidate tokens by the gradient surrogate is equivalent to ranking them at random, which is exactly the gradient fallacy of the previous section, now visible at the level of the proposal itself. The same measurement on the three GFlowNet variants yields correlations of 0.024, 0.057, and 0.046, all negligible, confirming that the phenomenon is not particular to the base model.

The mechanism behind this null result is the mismatch between the radius over which the linear approximation is valid and the size of the smallest move the sampler can make. A first-order Taylor expansion is accurate only within a small neighbourhood of the point at which it is taken; a Transformer [Vaswani et al., 2017], with its layer normalization and attention nonlinearities, is a violently nonlinear function of its input embeddings, so this neighbourhood is small. Figure 5.5 shows the correlation between the surrogate and the true change computed *within* bins of embedding distance. The correlation is at best weakly positive at the very smallest distances and decays to zero well before the mean inter-token distance of 2.35 is reached. But the mean inter-token distance is, by definition, the scale of the smallest possible substitution: every candidate token lies at least that far away. The sampler is therefore always evaluating its linear surrogate at distances far outside the radius over which that surrogate carries any signal. The gradient is not merely imprecise here; it is

The proposal surrogate does not predict the true energy change

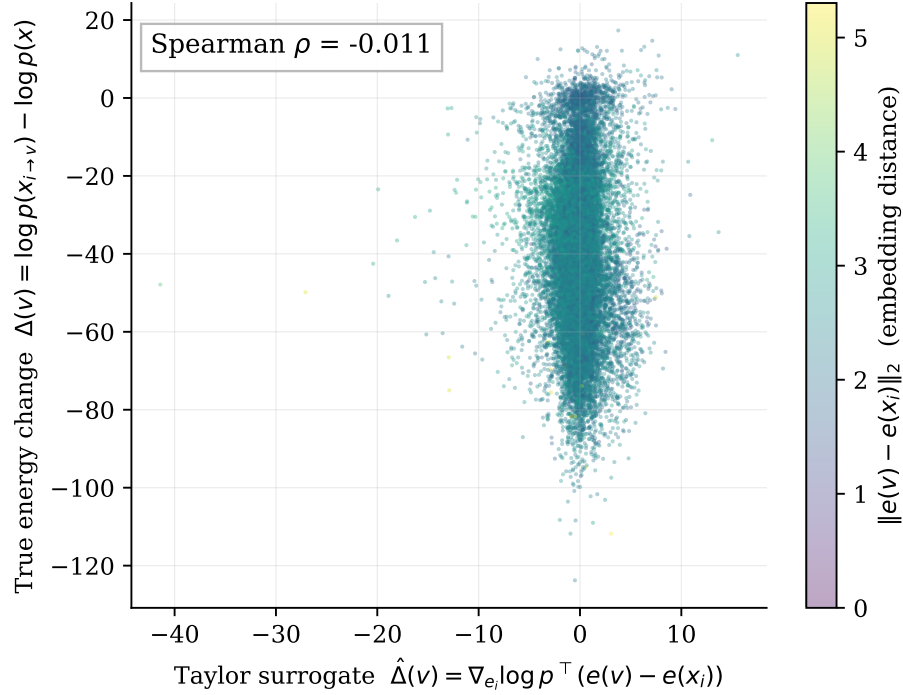


Figure 5.4: The first-order Taylor surrogate used by the discrete proposal, plotted against the true change in sequence log-likelihood, for GPT-2 Large. Points are coloured by the embedding distance of the candidate token. The two quantities are uncorrelated (Spearman $\rho = -0.011$), so ranking tokens by the surrogate is equivalent to ranking them at random.

inapplicable, because every admissible discrete step leaves the region in which the local linear model means anything.

A second, independent limitation compounds the first. The gradient is taken with respect to the input embedding of the masked position, and it can therefore only register the effect of a substitution on the *future* tokens, whose probabilities depend on that embedding through the network. It is structurally blind to the effect of the substitution on the probability of the candidate token *itself* given its left context, because that probability depends on the candidate through a discrete index into the output softmax, not through the input embedding. The decomposition in the linearization experiment measures both effects: the mean absolute magnitude of the self term is 15.0 nats, and the mean absolute magnitude of the future term is 24.2 nats, so the self term is far from negligible, and the surrogate correlates with the future term ($\rho = 0.03$) rather than the self term ($\rho = -0.10$). The gradient is thus blind to a term of substantial magnitude, which further degrades its value as a ranking signal. The full decomposition is shown in the appendix, Figure A.1, and the top- k recall analysis confirming that gradient ranking does not exceed chance is shown in Figure A.2. Together these answer the “why” behind RQ1 at the level of

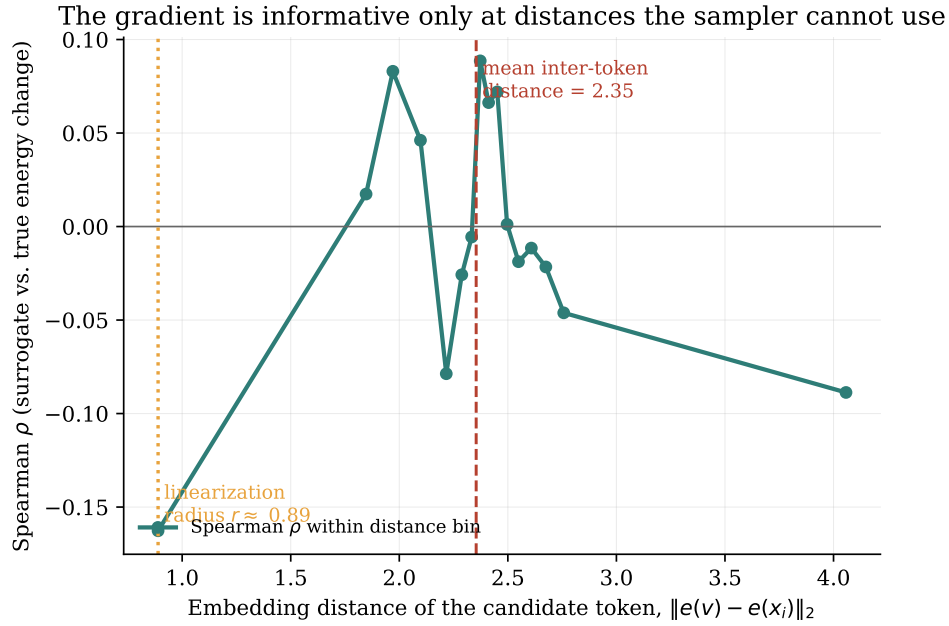


Figure 5.5: Correlation between the gradient surrogate and the true energy change, as a function of the embedding distance of the candidate token, for GPT-2 Large. The correlation decays to zero well within the mean inter-token distance (dashed line), so at the distances of any admissible token substitution the surrogate is uninformative.

the underlying mechanism.

5.6 The Likelihood Trap

The gradient fallacy explains why the samplers cannot move toward low energy effectively. The likelihood trap explains why doing so would not help even if they could. The whole energy-based framework rests on the premise that low energy, that is, high likelihood, corresponds to good text. On the study’s own models, it does not, reproducing on our specific models the neural-text-degeneration phenomenon of Holtzman et al. [2020]. The likelihood-trap experiment decodes with several strategies and measures the relationship between the per-token likelihood of the generated text and its repetition rate. Figure 5.6 shows the result for GPT-2 Large: as per-token likelihood increases, repetition increases and lexical diversity falls, monotonically. The correlation between per-token likelihood and repetition rate is +0.65 on the base model, and it is stronger still on the GFlowNet-tuned variants, reaching +0.93 on the length-normalized variant. Beam search, which most nearly approximates the mode of the distribution, produces the highest-likelihood and most degenerate text of all.

This has a sharp implication for the energy-based programme. Any procedure

The lowest-energy text is the worst text: the likelihood trap

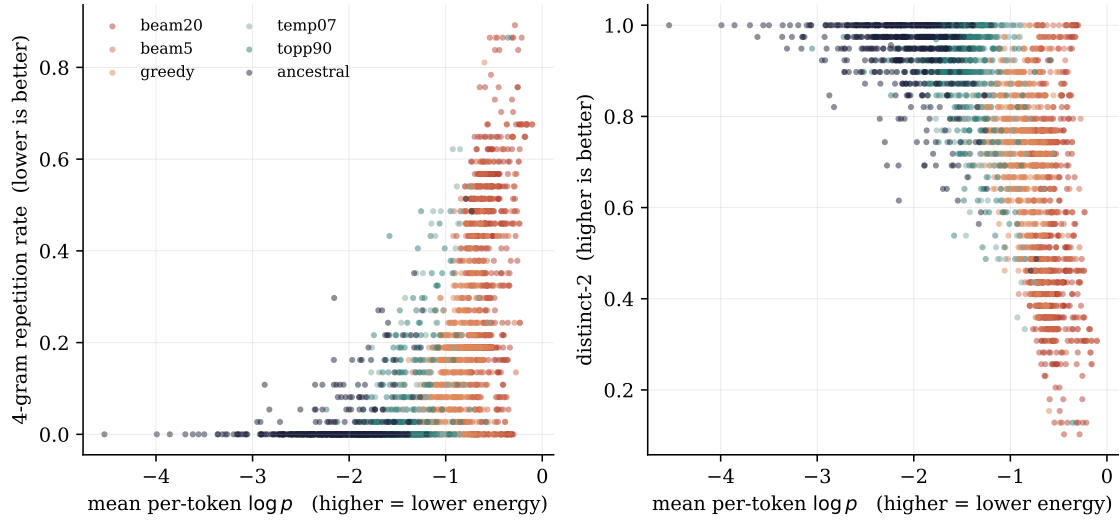


Figure 5.6: The likelihood trap on GPT-2 Large. As mean per-token log-probability increases (lower energy), the 4-gram repetition rate rises and the distinct-2 diversity falls. The highest-likelihood decoding strategies, greedy and beam search, produce the most degenerate text. Low energy is not good text.

that succeeds in concentrating probability mass on the lowest-energy regions of this energy, whether an annealed Langevin sampler in its quenching phase or a GFlowNet trained on the likelihood as a reward, is thereby concentrating it on degenerate text. Success at the stated optimization objective is failure at the underlying goal. The regression of total log-likelihood on generated length, computed over the sampled generations where length varies, has a slope of -1.12 nats per token, which quantifies a further consequence taken up in the GFlowNet section: because every additional token lowers the unnormalized likelihood, an energy defined as the raw log-likelihood contains an implicit and substantial incentive toward brevity. The length-versus-likelihood regression is shown in the appendix, Figure A.4.

5.7 Embedding Geometry and Anisotropy

The anisotropy analysis, referenced in earlier sections, is reported here in full because it underpins both the step-size result and the unreliability of Euclidean distance as a metric, and because the anisotropy of Transformer embeddings is itself a documented phenomenon [Gao et al., 2019, Ethayarajh, 2019]. In GPT-2 Large the mean pairwise distance between token embeddings is 2.77 and the mean nearest-neighbour distance is 1.82; in Llama-3 the corresponding figures are 0.84 and 0.59. The mean pairwise cosine similarity is well above zero in both models (0.086 and 0.018 respec-

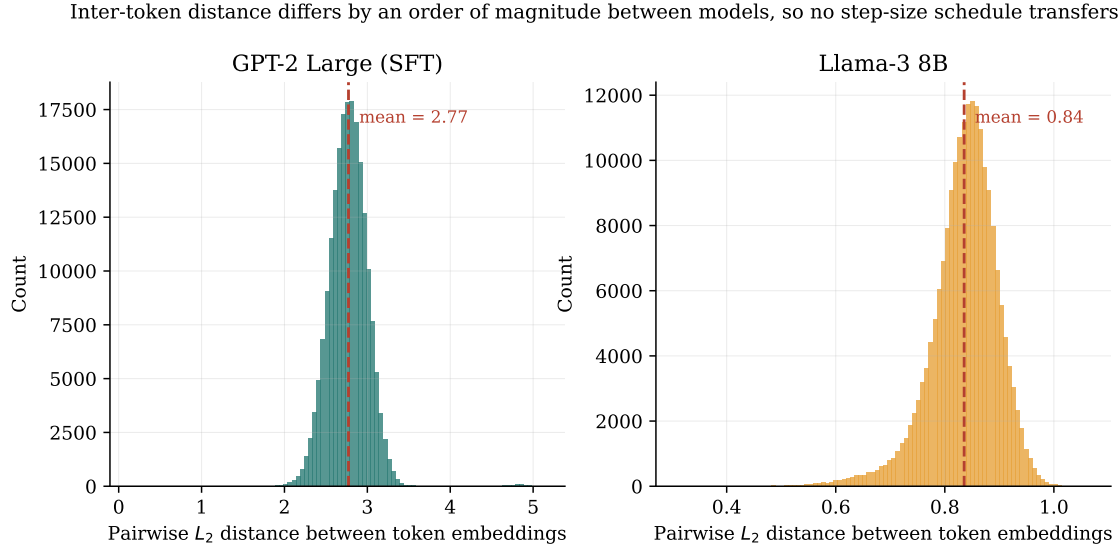


Figure 5.7: Distributions of pairwise token-embedding distances in GPT-2 Large and Llama-3, on separate scales. The mean inter-token distance differs by roughly a factor of three between the two models, which is why no single step-size schedule transfers between them.

tively), confirming the cone-shaped anisotropy documented in the literature. The top principal component accounts for only a small fraction of the variance, so the anisotropy is a broad property of the space rather than a single dominant direction. Figure 5.7 shows the distributions of pairwise distances for the two models on their separate scales. The order-of-magnitude difference in these distances is the direct cause of the failure of a Llama-tuned step size on GPT-2, and the compression of distances that anisotropy produces is why a token can be close in embedding space yet grammatically wrong, which is the reason the study relies on KL divergence rather than Euclidean distance as its primary metric.

5.8 GFlowNet Fine-Tuning

The results to this point concern the gradient-guided samplers. Since their failure traces to the gradient, the natural remedy is to abandon the gradient and amortize the search with a GFlowNet [Bengio et al., 2021, Hu et al., 2024], which only ever evaluates the reward. This section reports what happens when that is done, and the summary is that amortization does not escape the pathology; it merely relocates it. Three distinct failure modes were observed.

The first is **capacity collapse**. Initial attempts on the smaller GPT-2 base produced non-linguistic output regardless of the reward configuration, indicating that the policy lacked the capacity to represent the task at all; moving to GPT-2 Large,

the model of the reference implementation of Hu et al. [2024], resolved this and is assumed throughout the reported runs. The second is **length collapse**, and it follows directly from the likelihood trap. With the unnormalized reward (`len_beta` = 0), the reward is the raw log-likelihood, every additional token lowers it by around 1.12 nats as measured in Section 5.6, and the policy consequently learns to produce extremely short sequences. Under free-running ancestral sampling the length-collapse variants never emit an end-of-sequence token at all, running to the generation cap in every case, which is the extreme form of the same incentive: the policy has learned that stopping is costly under an unnormalized-likelihood reward. The length collapse is not a defect of the optimizer; it is the optimizer correctly following a reward with an implicit brevity incentive. The third is **reward hacking**. Normalizing the reward by length (`len_beta` = 1) removes the brevity incentive and restores the length, but the output ceases to be linguistic: the policy discovers sequences that achieve high length-normalized likelihood without being coherent text, producing strings of high-probability tokens with no grammatical structure. The two settings of the single length parameter thus produce two different degeneracies, and neither produces good text.

These three modes are unified by a single observation: the collapse is not fundamentally about length. Length is the visible symptom, but the underlying cause is that the reward, an autoregressive likelihood that was never shaped to serve as an energy over sequences, has no well-behaved optimum where coherent text lies, so every repair of one degeneracy surfaces another. The decisive evidence for locating the cause in the energy rather than the search is the unifying experiment. If GFlowNet training had reshaped the model’s likelihood into a smoother, more navigable landscape, then running the Langevin sampler on the GFlowNet-tuned model should behave differently from running it on the base model. It does not. Table 5.2 reports the final KL divergence of the discrete sampler on the base model and on the three GFlowNet variants, under identical conditions. The four values are within noise of one another, and the gradient fallacy replicates on every GFlowNet variant exactly as on the base. Amortizing the energy does not repair the landscape; the frozen-likelihood pathology survives retraining, which is the strongest possible evidence that it is a property of the energy and not of any sampler. This answers RQ3 directly: amortization does not escape the failure, because the failure is in the energy that both the sampler and the amortized policy inherit.

Energy function	Final KL (DLS, 50 steps, policy, MH, gn)
GPT-2 Large (SFT base)	6.541
GFlowNet, <code>len_beta</code> = 0, 500	6.306
GFlowNet, <code>len_beta</code> = 0, 2000	6.721
GFlowNet, <code>len_beta</code> = 1, 500	6.415

Table 5.2: Final KL divergence of the discrete Langevin sampler on the base model and on the three GFlowNet-tuned variants, under identical conditions. Source: `final_kl_by_model.csv`. The four values lie within noise of one another: Langevin sampling on the amortized energy is indistinguishable from sampling on the untuned base, so the amortization does not repair the energy landscape.

5.9 Constrained Generation

The final experiment tests the plug-and-play claim directly (RQ4) by adding a sentiment constraint to the energy and measuring whether generation can be steered, in the spirit of the gradient-based constrained samplers of Kumar et al. [2022] and Qin et al. [2022]. Table 5.3 reports the steering effect for the discrete sampler across the five constraint modes, on the infilling task with a positive-sentiment target. The results are small in magnitude and inconsistent in sign. The full combined energy produces a steering gain near zero; the constraint-only mode does not reliably exceed the random baseline; and no mode produces the systematic movement toward the target sentiment that a working constraint would require. This is the expected outcome given everything that precedes it: the constraint gradient is subject to the same discrete-manifold difficulties as the fluency gradient, and adding it to an energy landscape that is already unnavigable does not produce controllable generation. Inspection of the generated text confirms the picture, showing the same degeneracy observed throughout. The plug-and-play promise of (1.2), that an additive constraint can steer a frozen model at inference time, is not realized on this energy landscape, because its precondition, a navigable fluency energy, does not hold. This answers RQ4 in the negative and, importantly, does so for a reason that the preceding sections have already established rather than as an isolated failure.

Constraint mode	Steering gain	Final KL
lm_only	0.0	8.558
cons_only	−8.0	8.308
full	−1.0	8.541
random	−7.0	8.512
cons_random	+1.0	8.528

Table 5.3: Steering effect of the sentiment constraint under the discrete sampler on the infilling task, positive-sentiment target. Source: `results_constrained/gpt2-large.infill1.ours.dls.*.tbl1.json`. The steering gain is the change in the percentage of generations classified as the target sentiment relative to the unconstrained baseline. The effects are small and inconsistent in sign; no mode produces reliable steering toward the target.

Chapter 6

Discussion

The results of the previous chapter are a collection of specific findings: a step size that does not transfer, an annealing artifact, a Metropolis–Hastings breakdown, a gradient that carries no information, a likelihood trap, three GFlowNet failure modes, and a constraint that does not steer. This chapter draws them together into a single account, states explicitly which research question each part of the results answers and where, connects the findings back to the related work of Chapter 3, considers what the account does and does not license us to conclude, and sets out the experiment that would falsify it.

6.1 Answers to the Research Questions

It is useful to state plainly which research question each result answers and in which section of Chapter 5 the answer is established.

RQ1, whether the frozen likelihood can be sampled effectively with faithful Langevin dynamics on the discrete manifold and if not why, is answered in the negative across Sections 5.1, 5.4, and 5.5. Section 5.1 shows that the sampler cannot even be made to move without model-specific calibration; Section 5.4 shows, through the direction-versus-magnitude ablation, that following the true gradient is indistinguishable from following a random direction of equal norm; and Section 5.5 supplies the mechanism, namely that the linearization radius of the surrogate is smaller than the smallest admissible discrete step, so the gradient is inapplicable rather than merely imprecise.

RQ2, the role of the Metropolis–Hastings correction in each sampler, is answered in Sections 5.2 and 5.3. For the discrete sampler the correction is beneficial and

removes the quenching artifact (Section 5.2); for the continuous sampler the same correction is catastrophic, and Section 5.3 attributes the collapse specifically to the proposal term, connecting it to the non-Lipschitz drift induced by nearest-neighbour projection. The correction does not make the continuous sampler work; it reveals that the continuous sampler was only ever functioning as a biased optimizer.

RQ3, whether amortization with a GFlowNet escapes the failure, is answered in Section 5.8. It does not: three distinct failure modes appear, and the unifying experiment shows that Langevin sampling on the amortized energy is indistinguishable from sampling on the untuned base, which locates the cause in the energy rather than the search.

RQ4, whether an additive constraint can steer generation on this landscape, is answered in Section 5.9. It cannot: the steering gains are small and inconsistent in sign, and the failure follows from the same defect established for the fluency energy.

6.2 A Single Defect, Reached by Two Routes

The organizing claim of this thesis is that all of the failures above are surface manifestations of one underlying defect: the frozen likelihood of an autoregressive language model is not a usable energy function for sampling, because the training objective that produces it never shapes it to be one. Teacher forcing optimizes each local conditional $p_{\text{LM}}(x_t \mid x_{<t})$ to predict the next token given a correct prefix. It places no constraint on how the joint log-likelihood behaves as a function of the sequence when the sequence is varied, and in particular it does not make that function smooth, or well-calibrated as a measure of quality, or navigable by local gradient information. Energy-based decoding borrows this quantity and treats it as though it had those properties. It does not.

The strength of the evidence for this claim rests on the fact that it is reached by two independent routes that share no machinery. The gradient-guided route differentiates the likelihood and finds that the gradient is uninformative (Section 5.4), that its surrogate is uncorrelated with the true energy change because the linearization radius is smaller than the smallest discrete step (Section 5.5), and that the Metropolis–Hastings correction, applied honestly, rejects every useful move because the projected drift is discontinuous (Section 5.3). The amortized route never touches the gradient at all; it treats the likelihood only as a scalar reward, and finds that the reward has no well-behaved optimum where coherent text lies, so the learned policy collapses in whichever direction the reward’s implicit incentives push it (Section 5.8).

That two methods with nothing in common beyond the energy they use should fail in ways that trace back to the same property of that energy is the central argument that the energy, and not the sampler, is the problem. The unifying experiment, in which Langevin sampling on the GFlowNet-tuned energy is indistinguishable from Langevin sampling on the base, closes the loop by showing that changing the model through amortization does not change the sampler’s fate.

The likelihood trap ties the two routes together at the level of intuition. Because low energy is degenerate text, the goal that both methods pursue is the wrong goal. The gradient-guided samplers cannot reach it, and that is fortunate, because reaching it, as the quenching phase and the GFlowNet reward hacking both show, produces bad text. The methods are, in a sense, protected from their objective by their inability to optimize it, and the moments when they do optimize it effectively are exactly the moments when their output degenerates.

6.3 Connections to the Related Work

The findings of this thesis stand in a definite relationship to the literature reviewed in Chapter 3, and it is worth making that relationship explicit rather than leaving it implicit.

The likelihood trap measured in Section 5.6 is a direct, quantitative reproduction on our own models of the neural-text-degeneration phenomenon that Holtzman et al. [2020] identified: that the most probable sequences under a neural language model are degenerate and that maximization-based decoding produces worse text than sampling. What Chapter 3 presented as a known hazard of decoding, this thesis re-derives as a structural obstacle for energy-based sampling specifically, because it means the target of the sampling procedure is itself undesirable. Our contribution is to connect that degeneration to the sampling objective and to measure the monotone likelihood-repetition relationship directly on the models under study.

The anisotropy results of Section 5.7 confirm, on GPT-2 Large and Llama-3, the representation-degeneration and embedding-anisotropy phenomena documented by Gao et al. [2019] and Ethayarajh [2019]. Chapter 3 raised anisotropy as a reason to distrust Euclidean distance in embedding space; the measurements here quantify that distrust and, more importantly, show a consequence those works did not consider, namely that the anisotropy scale differs enough between models to make a single step-size schedule non-transferable, which is the concrete cause of the calibration failure in Section 5.1.

The Metropolis–Hastings breakdown of Section 5.3 is the empirical counterpart to a gap noted in Section 3.2: that the energy-based decoding literature frequently omits the correction, and where it is included, the difficulty of computing the reverse proposal on a projected landscape is seldom examined. This thesis takes the correction seriously, implements it faithfully following the convergence theory of Roberts and Tweedie [1996], and measures exactly what the omission was hiding. Methods such as COLD [Qin et al., 2022] and MuCoLa [Kumar et al., 2022], which our continuous sampler follows, are in this light best understood as biased optimizers rather than samplers, and their reported successes are successes of optimization under early stopping, not of sampling from the stated energy.

Finally, the GFlowNet results of Section 5.8 qualify the picture presented by Hu et al. [2024], whose formulation and codebase this thesis builds on. Their work showed that amortized sampling from a frozen-model posterior can yield diverse, transferable samples on tasks where the reward is well-behaved. Our narrower and more critical finding is that when the reward is the raw frozen likelihood used as an energy over free-form sequences, the amortization inherits the pathology of that energy, and the learned policy collapses. The two findings are compatible: the difference lies in whether the reward defines a landscape with a coherent-text optimum, and our contribution is to show that the frozen autoregressive likelihood does not.

6.4 What the Results Do and Do Not Show

It is important to state the limits of these conclusions precisely. The thesis shows that the frozen likelihood of a *standard autoregressive* language model is a poor energy for gradient-guided and amortized sampling on discrete text, and it explains why in terms of the training objective and the geometry of the token embedding space. It does not show that energy-based controllable generation is impossible in general. It shows that a particular and widely used instantiation of it fails, and it identifies the property responsible.

Several caveats bound the empirical claims. The KL-divergence metric depends on the position of the corrupted token, which is why the central results are argued from the absence of a gap between conditions rather than from small measured differences; a null gap is robust to a noisy metric in a way that a narrow win would not be. The Llama-3 results support only qualitative cross-architecture claims, because the tokenizer and embedding scale differ from the GPT-2 family. The linearization result is that the surrogate is uniformly uninformative at all admissible distances

rather than informative up to a sharp radius and useless beyond it; the language of a linearization radius is a useful summary, but the measured correlation is low everywhere the sampler operates rather than crossing a clean threshold. And the constrained-generation extension establishes the absence of reliable steering on this energy, not the impossibility of steering on a better one.

The thesis is also candid about the relationship between its title and its content. The original objective was controllable generation via faithful Langevin dynamics, realized as a two-stage plan in which the first stage would establish that energy-guided sampling on a frozen model is viable and the second would add the constraint term and perform sentiment and toxicity control. The first stage did not hold. Rather than proceed to build control on a foundation the study had shown to be unsound, which would have produced results without interpretable meaning, the work turned to diagnosing why the foundation fails. What it delivers is therefore a systematic diagnosis of the failure of a prerequisite, with the mechanisms identified and the constrained-generation extension included as direct evidence, rather than a demonstration of successful control.

6.5 Implications and the Falsifying Experiment

The most valuable consequence of locating the defect in the training objective is that it makes a sharp, testable prediction. If the reason the frozen autoregressive gradient is useless is that teacher forcing never shapes a score function over sequences, then a model whose training objective *does* shape such a function should not suffer the same fate. **Diffusion language models** are exactly such models [Li et al., 2022, Lou et al., 2024]. A diffusion model is trained, by construction, to denoise corrupted inputs, and denoising is equivalent to learning the gradient of the log-density of the data, the score function, by the identity of Vincent [2011] that also underpins score-based generative modeling [Song and Ermon, 2019, Ho et al., 2020]. Running the identical protocol of this thesis on a diffusion language model therefore constitutes a positive control. If Langevin sampling succeeds there, where it fails on the autoregressive model, the training objective is isolated as the causal variable and the diagnosis of this thesis is confirmed. If it fails there too, the diagnosis is wrong and a deeper cause must be sought. Either outcome is informative, and the experiment is inexpensive because score-based discrete diffusion models sharing the GPT-2 tokenizer are available. This is, in the author’s view, the single most important next step, and the design of this thesis was shaped to make it a clean test.

A second and complementary direction is to attack the defect at its source rather than route around it. If the problem is that the likelihood is not a score function, one option is to fine-tune the model so that its sequence log-density is aligned directly with a target energy, using a score-matching objective, which is cheaper than the reinforcement-learning-style training of a GFlowNet and directed precisely at the identified deficiency. Only once an energy landscape is known to be navigable does the additive constraint term of the plug-and-play framework become meaningful, and at that point the original goal of controllable generation becomes reachable. The contribution of this thesis to that longer programme is a map of where the ground gives way, and a falsifiable hypothesis about why.

Chapter 7

Conclusion

This thesis set out to realize the promise of energy-based controllable text generation, in which a frozen autoregressive language model is treated as an energy function and steered at inference time by sampling from it with gradient-guided Markov chain Monte Carlo. Across a study of one hundred and forty-five sampling configurations, five energy functions, and two samplers, it found that the central premise of the framework does not hold: the frozen likelihood of an autoregressive language model is not a usable energy function for sampling on discrete text.

The contribution of the thesis is not this negative observation alone but the mechanistic account that accompanies it, established by independent means and supported at every step by direct measurement. Following the true gradient was shown to be statistically indistinguishable from following a random direction of equal magnitude, on all five energy functions, once a normalization confound was removed by an ablation designed for the purpose. This gradient fallacy was explained by a linearization failure: the first-order surrogate that the discrete sampler uses to rank candidate tokens is uncorrelated with the true change in energy, because the smallest admissible discrete step is far larger than the radius over which the local linear model is valid. The Metropolis–Hastings correction, applied faithfully, was shown to paralyze the continuous sampler, and the trace analysis located the cause precisely in the proposal term, which collapses because the nearest-neighbour projection makes the drift discontinuous at every cell boundary. A likelihood trap was measured on the study’s own models, showing that the lowest-energy text is the most degenerate, so that success at the optimization objective is failure at the underlying goal. Turning to amortized inference, the GFlowNet experiments exhibited three distinct failure modes, and a unifying experiment showed that Langevin sampling on the GFlowNet-tuned energy is indistinguishable from sampling on the untuned base, demonstrating

that the pathology survives amortization and is therefore a property of the energy rather than of any sampler. A constrained-generation extension confirmed that the constraint gradient carries no reliable steering signal on this landscape.

Taken together, these findings locate the cause of the failure in the training objective of autoregressive language models, which shapes accurate local conditional distributions and never a smooth global energy over sequences. This diagnosis makes a falsifiable prediction, that a diffusion language model whose objective does shape a score function should not fail in the same way, and the thesis sets out that positive-control experiment as the natural next step. What the thesis provides to the larger effort of building inference-time controllable generation on frozen models is therefore a precise characterization of where and why the most common approach breaks down, together with an experiment capable of confirming or refuting the explanation. A negative result about a method, when it is accompanied by a mechanism, a measurement, and a falsifiable hypothesis, is a positive result about understanding, and it is in that sense that the work is offered.

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Appendix A

Appendix

A.1 Additional Diagnostic Figures

This section collects supporting figures referenced in Chapter 5. Each figure is discussed at the point of reference in the results chapter, as indicated in the captions below.

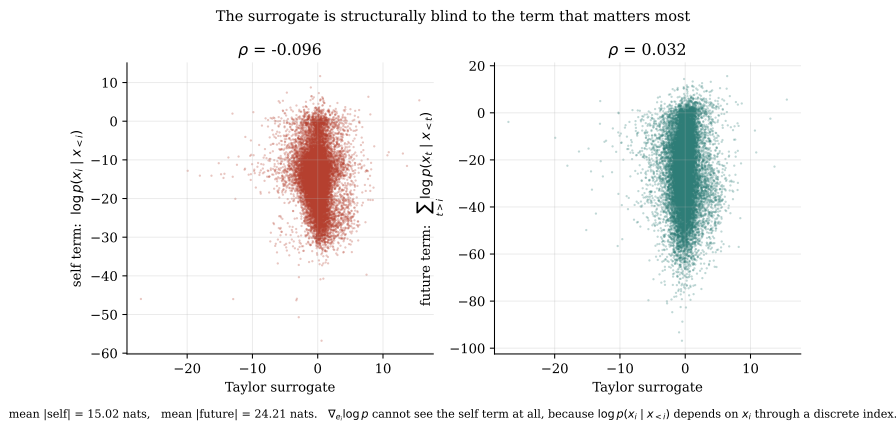


Figure A.1: Decomposition of the true energy change into the self term and the future term, plotted against the gradient surrogate, for GPT-2 Large. The surrogate correlates weakly with the future term and is anti-correlated with the self term, to which it is structurally blind. This figure supports the argument of Section 5.5.

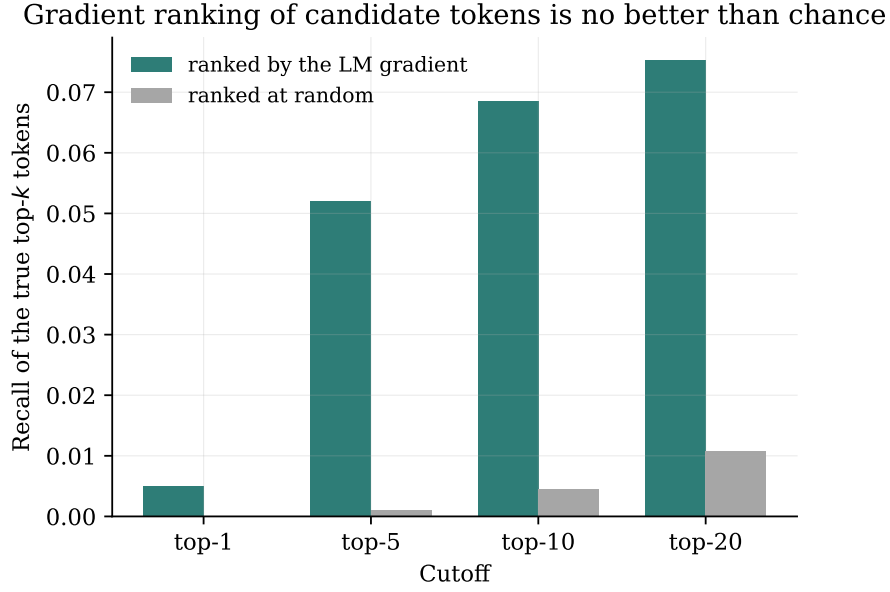


Figure A.2: Recall of the true top- k token substitutions by the gradient-ranked top- k , against a random-ranking baseline, for GPT-2 Large. Gradient ranking does not exceed chance, corroborating the null correlation reported in Section 5.5.

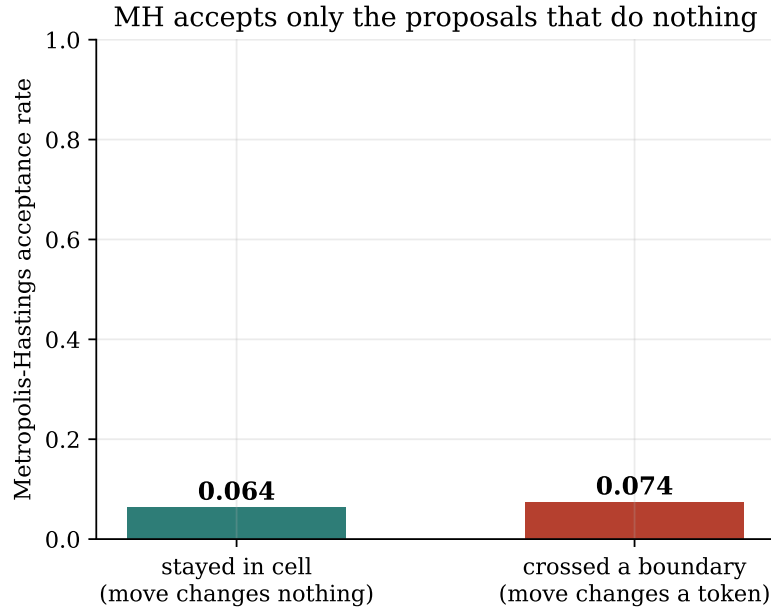


Figure A.3: Metropolis-Hastings acceptance rate in the continuous sampler, conditioned on whether the proposal crossed a Voronoi cell boundary. Moves that change a token are almost never accepted. This figure supports Section 5.3.

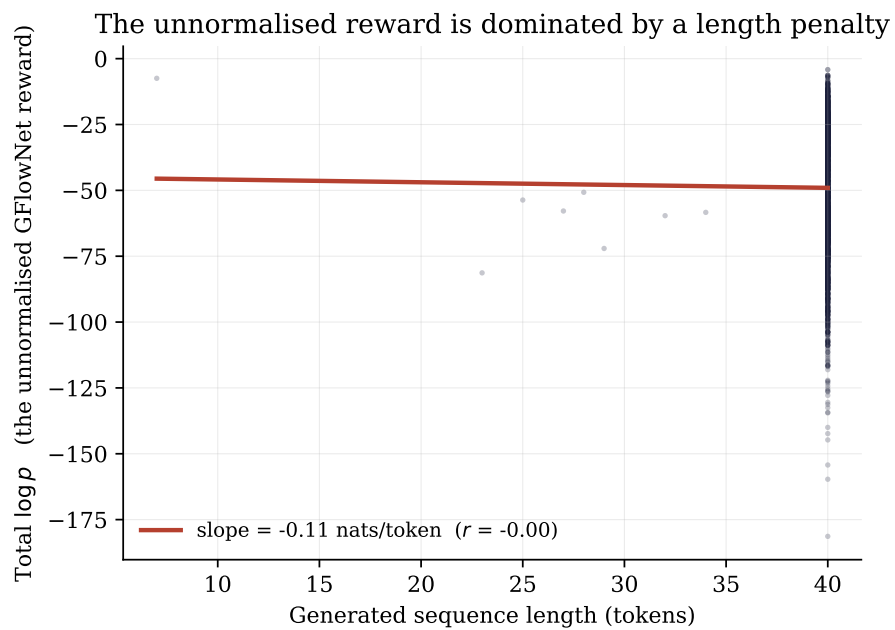


Figure A.4: Total log-likelihood against generated length for GPT-2 Large. The negative slope, approximately -1.12 nats per token, is the implicit brevity incentive in an unnormalized-likelihood reward, discussed in Section 5.6 and Section 5.8.

A.2 Full Configuration Grid

Table A.1 reports the final KL divergence for a representative subset of the full configuration grid across the four GPT-2-family energy functions, drawn from the aggregated results file `final_kl_by_model.csv`. The complete grid comprises the factorial combination of sampler (DLS, CLS), proposal method (policy, grad-norm-preserved, random), Metropolis–Hastings correction (on, off), gradient normalization (on, off), and schedule length (50, 100), evaluated on each of the five energy functions, for a total of 145 configurations.

Configuration	GPT-2 Large	GFN lb0-500	GFN lb0-2000	GFN lb1-500
DLS policy, MH, gn, s50	6.541	6.306	6.721	6.415
DLS policy, no-MH, gn, s50	9.499	9.195	10.085	8.669
DLS random, MH, gn, s50	6.370	6.105	6.548	5.525
DLS gnp-random, MH, gn, s50	6.370	6.105	6.548	5.525
DLS policy, MH, no-gn, s50	6.474	6.009	6.698	6.008
DLS gnp-random, MH, no-gn, s50	6.430	6.001	6.552	5.338
CLS policy, MH, no-gn, s50	8.083	7.799	8.198	13.732
CLS policy, no-MH, no-gn, s50	8.083	7.799	8.198	13.732

Table A.1: Representative subset of the final-KL configuration grid, from `final_kl_by_model.csv`. Note that the DLS policy and grad-norm-preserved random rows are near-identical under both normalization settings, which is the gradient fallacy of Section 5.4. Note also that the CLS rows with and without the Metropolis–Hastings correction are identical to three decimals, because under the correction the sampler rejects essentially all proposals and does not move, as established in Section 5.3.

A.3 Use of AI-Tools

The author of this thesis has used AI tools, i.e. Google AI Studio (Gemini 3.1 Pro) and Claude, in order to assist in the code support when errors were encountered in the implementation of the sampling and evaluation code and in order to secure their compliance with their accompanied papers. These tools were used in particular to help write and debug the job-scheduling and GPU-usage scripts that distribute the experiments across the available hardware, and the diagnostic experiment and plotting scripts. AI was also used for the analysis part of the thesis, again with support in the implementation of the analysis code and the encountered errors, and in some cases in order to compile the tables seen in this thesis and to refine the text. All the other parts of the thesis, including the writing, the background work, the related work, the experimental design, the analysis and the interpretation of the results, were done solely by the author, who has verified all quantitative claims against the underlying experimental output and assumes full responsibility for the content of this work.